HALF YEARLY EXAMINATION - 2025

12 STD. BIOLOGY

TIME : 3 Hrs MARKS : 70

Bio Botany Marks : 35

I Answer All the questions. 8x1=8

1. A Plant called x possesses small flower with reduced perianth and versatile anther. The probable agent for pollination would be
   a) Water (b) Wind c) butterflies d) beetles
2. "Gametes are never hybrid". This is a statement of
   a) Law of dominance b) Law of independent assortment (c) Law of segregation
   d) Law of random fertilization.
3. In which techniques Ethidium Bromide is used.?
   a) Southern Blotting techniques b)Western Blotting techniques
   c) Polymerase Chain Reaction (d) Agarose Gel Electroporosis
4. Virus free plants are developed from
   a) Organ culture (b) Meristem culture c) Protoplast culture d) Cell suspension culture
5. In soil water available for plants is
   a) gravitational water b) chemically bound water (c) capillary water d) hygroscopic water
6. Which of the following is not a sedimentary cycle
   (a) Nitrogen cycle b) Phosphorous cycle c) Sulphur cycle d) Calcium cycle
7. Depletion of which gas in the atmosphere can lead to an increased incidence of skin cancer.
   a) Ammonia b) Methane c) Nitrous oxide (d) Ozone
8. The active principle trans-tetra hydro canabial is present in
   a) Opium b) Curcuma (c) Marijuana d) Andrographis

II Answer any FOUR questions. 4x2=8

9. Write short note on Heterostyly. 14
10. What is back cross? 37 / 49
11. What is Crossing over? 54
12. Name the chemicals used in gene transfer? 76
13. What is vivipary? Name the plant group which exhibits vivipary. 127 133
14. Enumerate the uses of turmeric. 192

III Answer any THREE questions. Qn. no 19 is Compulsory. 3x3=9

15. Write the effect of Global warming in plants. 157
16. How is rhytidome act as the structural defence by plants against fire. 115
17. Write the benefits and risk of Genetically Modified foods. 85
18. What is Gene mapping? Write its uses. 56
19. Draw and label the parts of monocot seed. 24

IV Answer the all questions. 2x5=10

20. a) What is endosperm.? Explain their types. *(OR)* 21
    b) Describe dominant epistasis with an example. 42
21. a) What are the different types of hybridization? Explain. *(OR)* 177
    b) How will you prepare an organic Pesticide for your home garden with the vegetables available from your kitchen? 200

TIRUVANNAMALAI DISTRICT
HALF YEARLY EXAMINATION 2025
BIO-BOTANY ANSWER KEY- 2025
PREPARED BY: ETS - SRIJAYAM ACADEMY

I.Answer All the Questions: **8x1=8**

1.b.Wind

2.c.Law of segregation

3.d.Agarose Gel Electroporosis

4.b.Meristem Culture

5.c.Capillary Water

6.a.Nitrogen Cycle

7.d.Ozone

8.c.marijuana

II.Answer any FOUR Questions: **4X2=8**

9.Write Short note on Heterostyly.

Some plants produce two or three different forms of flowers that are different in their length of stamens and style. Pollination will take place only between organs of the same length.

10.What is Back cross?

Crosses between F1 off-springs with either of the two parents (hybrid) are known as back cross.

11.What is crossing over?

Crossing over is a biological process that produces new combination of genes by inter-changing the corresponding segments between non-sister chromatids of homologous pair of chromosomes. The term 'crossing over' was coined by **Morgan (1912).**

12.Name the chemicals used for Gene transfer.

Chemical mediated gene transfer: Certain chemicals like polyethylene glycol (PEG) and dextran sulphate induce DNA uptake into plant protoplasts.

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13.What is vivipary? Name the plant group which exhibits vivipary.

Seeds germinate in the fruits of mother plant itself (**Vivipary).**

Vivipary: When seeds or embryos begin to develop before they detach from the parent.

14.Enumerate the uses of turmeric.

Uses

1. Turmeric is one of the most important and ancient Indian spices and used traditionally over thousands of years for culinary, cosmetic, dyeing and for medicinal purposes.
2. It is an important constituent of curry powders.
3. Turmeric is used as a colouring agent in pharmacy, confectionery and food industry.
4. Rice coloured with turmeric (yellow) is considered sacred and auspicious which is used in ceremonies. It is also used for dyeing leather, fibre, paper and toys.
5. Curcumin extracted from turmeric is responsible for the yellow colour. Curcumin is a very good anti-oxidant which may help fight various kinds of cancer.
6. It has anti-inflammatory, anti-diabetic, anti-bacterial, anti-fungal and anti-viral activities. It stops platelets from clotting in arteries, which leads to heart attack.

III.Answer any THREE questions.Qn.no 19 is Compulsary. 3X3=9

15.Write the effects of Global warming in plants.

1. Low agricultural productivity in tropics
2. Frequent heat waves (Weeds, pests, fungi need warmer temperature)
3. Increase of vectors and epidemics
4. Strong storms and intense flood damage
5. Water crisis and decreased irrigation
6. Change in fl owering seasons and pollinators
7. Change in Species distributional ranges
8. Species extinction.

16.How is rhytidome act as the structural defence by plants against fire.

1.It is the structural defense by plants against fire .
2.The outer bark of trees which extends to the last formed periderm is called Rhytidome.

3.It is composed of multiple layers of suberized periderm, cortical and phloem tissues.
4.It protects the stem against fire , water loss, invasion of insects and prevents infections by microorganisms.

17.Write the benefits and risk of genetically modified foods.

GM Food - Benefits

1. High yield without pest
2. 70% reduction of pesticide usage
3. Reduce soil pollution problem
4. Conserve microbial population in soil

Risks - believed to

5. Affect liver, kidney function and cancer
6. Hormonal imbalance and physical disorder
7. Anaphylactic shock (sudden hypersensitive reaction) and allergies.
8. Adverse effect in immune system because of bacterial protein.
9. Loss of viability of seeds show in terminator seed technology of GM crops.

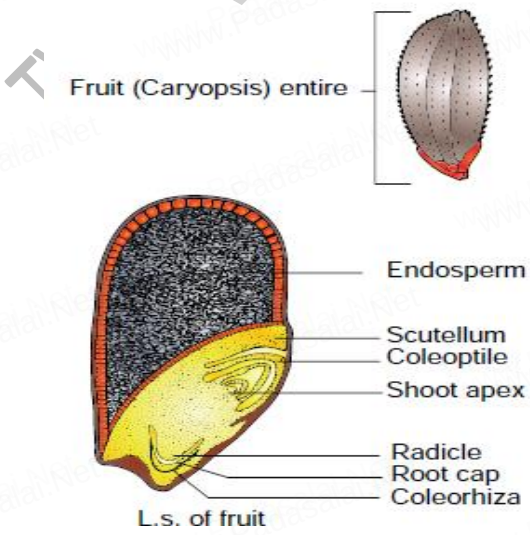
18.What is Gene Mapping? Write its uses.

The diagrammatic representation of position of genes and related distances between the adjacent genes is called **genetic mapping**.

Uses of genetic mapping

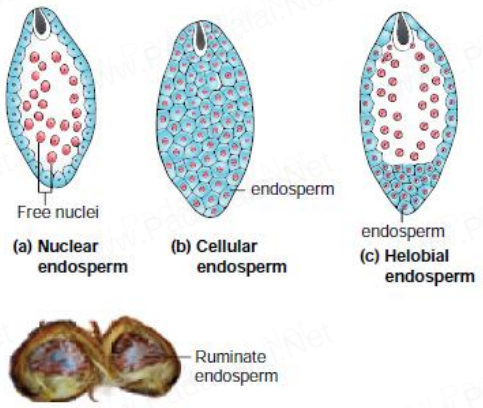
1. It is used to determine gene order, identify the locus of a gene and calculate the distances between genes.
2. They are useful in predicting results of dihybrid and trihybrid crosses.
3. It allows the geneticists to understand the overall genetic complexity of particular organism.

19.Draw and label the parts of monocot seed.



IV.Answer all the Questions: **2X5=10**

20. a) What is endosperm? Explain their types. (OR)



Endosperm

1. The primary endosperm nucleus (PEN) divides immediately after fertilization but before the zygote starts to divide, into an endosperm.
2. The primary endosperm nucleus is the result of triple fusion (two polar nuclei and one sperm nucleus) and thus has 3n number of chromosomes.
3. It is a nutritive tissue and regulatory structure that nourishes the developing embryo. Depending upon the mode of development three types of endosperm are recognized in angiosperms.
4. They are nuclear endosperm, cellular endosperm and helobial endosperm

1.Nuclear endosperm: Primary Endosperm Nucleus undergoes several mitotic divisions without cell wall formation thus a free nuclear condition exists in the endosperm. Examples: *Coccinia*, *Capsella* and *Arachis*

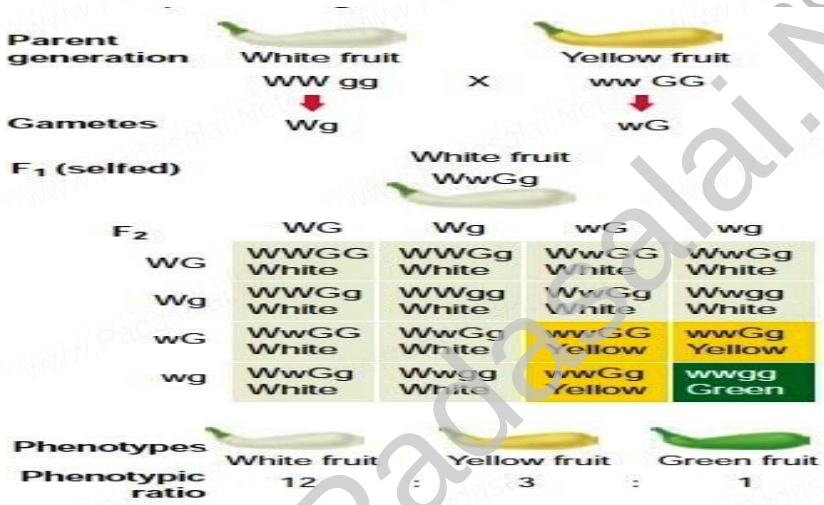
2.Cellular endosperm: Primary endosperm nucleus divides into 2 nuclei and it is immediately followed by wall formation. Subsequent divisions also follow cell wall formation. Examples: *Adoxa, Helianthus* and *Scoparia*

1. **3.Helobial endosperm:** Primary Endosperm Nucleus moves towards base of embryo sac and divides into two nuclei.
2. Cell wall formation takes place leading to the formation of a large micropylar and small chalazal chamber.
3. The nucleus of the micropylar chamber undergoes several free nuclear division whereas that of chalazal chamber may or may not divide.
4. Examples : *Hydrilla* and *Vallisneria*.

5. The endosperms may either be completely consumed by the developing embryo or it may persist in the mature seeds.
6. Those seeds without endosperms are called non- endospermous or ex-albuminous seeds. Examples: Pea, Groundnut and Beans.
7. Those seeds with endosperms are called endospermous or albuminous seeds.
8. The endosperms in these seeds supply nutrition to the embryo during seed germination. Examples: Paddy, Coconut and Castor.

4.Ruminate endosperm: The endosperm with irregularity and unevenness in its surface forms ruminate endosperm.
Examples :*Areca catechu, Passiflora* and *Myristica*

20.b) Describe the dominant epistasis with an example.



Dominant epistasis in summer squash

1. **Dominant Epistasis –** It is a gene interaction in which two alleles of a gene at one locus interfere and suppress or mask the phenotypic expression of a different pair of alleles of another gene at another locus.
2. The gene that suppresses or masks the phenotypic expression of a gene at another locus is known as **epistatic**.
3. The gene whose expression is interfered by non-allelic genes and prevents from exhibiting its character is known as **hypostatic**.
4. When both the genes are present together, the phenotype is determined the epistatic gene and not by the hypostatic gene.
5. In the summer squash the fruit colour locus has a dominant allele 'W' for white colour and a recessive allele 'w' for coloured fruit.
6. 'W' allele is dominant that masks the expression of any colour. In another locus hypostatic allele 'G' is for yellow fruit and its recessive allele 'g' for green fruit.
7. In the first locus the white is dominant to colour where as in the second locus yellow is dominant to green.
8. When the white fruit with genotype WWgg is crossed with yellow fruit with genotype wwGG, the F1 plants have white fruit and are heterozygous

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(WwGg). When F1 heterozygous plants are crossed they give rise to F2 with the phenotypic ratio of 12 white : 3 yellow : 1 green.

9. Since W is epistatic to the alleles 'G' and 'g', the white which is dominant, masks the effect of yellow or green.
10. Homozygous recessive ww genotypes only can give the coloured fruits (4/16). Double recessive 'wwgg' will give green fruit (1/16).
11. The Plants having only 'G' in its genotype (wwGg or wwGG) will give the yellow fruit(3/16).

21.a)What are the different types of Hybridization ? Explain. (OR)

Types of Hybridization According to the relationship between plants, the hybridization is divided into.

i. **Intravarietal hybridization** - The cross between the plants of same variety. Such crosses are useful only in the self-pollinated crops.

ii. **Intervarietal hybridization -** The cross between the plants belonging to two diff erent varieties of the same species and is also known as intraspecifi c hybridization. Th is technique has been the basis of improving self-pollinated as well as cross pollinated crops

iii. **Interspecifi c hybridization -** The cross between the plants belonging to diff erent species belonging to the same genus is also called intragenic hybridization. It is commonly used for transferring the genes of disease, insect, pest and drought resistance from one species to another.
Example: *Gossypium hirsutum* x *Gossypium arboreum* – Deviraj.

iv. Intergeneric hybridization – The crosses are made between the plants belonging to two diff erent genera. The disadvantages are hybrid sterility, time consuming and expensive procedure. **Example:** Raphanobrassica, Triticale.

21.b)How will you prepare an organic Pesticide for your home garden with the vegetables available from your kitchen?

1. Mix 120g of hot chillies with 110 g of garlic or onion. Chop them thoroughly

2. Blend the vegetables together manually or using an electric grinder until it forms a thick paste

3. Add the vegetable paste to 500 ml of warm water. Give the ingredients a stir to thoroughly mix them together.

4. Pour the solution into a glass container and leave it undisturbed for 24 hours. If possible, keep the container in a sunny location. If not, at least keep the mixture in a warm place.

5. Strain the mixture. Pour the solution through a strainer, remove the vegetables and collect the vegetable-infused water and pour into another container. This filtrate is the pesticide. Either discard the vegetables or use it as a compost.

6. Pour the pesticide into a squirt bottle. Make sure that the spray bottle has first been cleaned with warm water and soap to get rid it of any potential contaminants. Use a funnel to transfer the liquid into the squirt bottle and replace the nozzle.

7. Spray your plants with the pesticide. Treat the infected plants every 4 to 5 days with the solution. After 3 or 4 treatments, the pest will be eliminated. If the area is thoroughly covered with the solution, this pesticide should keep bugs away for the rest of the season.

8. Avoid spraying the plants during the sunny times of the day since it could burn plants. Many other plants possess insect repellent or insecticidal properties. Combinations of these plants can be fermented and used as biopesticide.

ALL THE VERY BEST

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